Project DB Model and Design Report

FitHub (Group 11)

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Functional Requirements

Users:

Members

- Can register and manage their own account and personal profile
- Can view current membership status, details, and attendance history
- Can enroll in or cancel class enrollments based on eligibility and class capacity

Staff

- Can create, update, or deactivate membership plans
- Can add, update, and manage member accounts, including walk-in registrations and freezing/reactivating memberships
- Can record and manage payments and view payment history
- Can manage classes by creating sessions, assigning trainers, and updating capacity
- Can check-in members and verify active membership status

Administrators

- Can perform all staff functions and access all system records
- Can reset staff passwords, change staff roles, and delete staff accounts
- Can override membership statuses

Functions:

Register Account

- Gym members can create a unique account by providing required information.
- Information includes first name, last name, email, and password.
- The system will store the password as a secure hash in the database.
- Registration failures will prompt an error message if the username or email already exists.

Account Login and Logout

- Registered members and staff must log into the system using verified credentials (username and password).
- Upon successful authentication, the user gains access to their respective dashboard (member or administrative).

- The system allows users to securely log out of their active sessions.

Delete Account

- Members have the option to permanently delete their account.
- The system will remove the member's personal information from the database, deactivate associated membership records, and invalidate active sessions.

Manage Memberships

- Logged in members can view their current membership plan, start/end dates, and status (Active, Frozen, Expired).
- Authorized staff can add new membership plans, update pricing or duration, and deactivate outdated plans.

Class Enrollment

- Members may enroll in an available class through the schedule interface.
- The system verifies active membership, checks that the class has not reached maximum capacity, and updates current capacity upon enrollment.
- If full, the member may be placed on a waitlist.
- Members may cancel enrollment before the scheduled class, which adjusts the capacity accordingly.

Manage Classes and Trainers

- Authorized staff can create new classes, assign trainers, modify schedule times, update maximum capacity, and cancel classes.
- Staff with managerial privileges can add new trainers, update trainer certifications or specializations, and remove trainer records.

Process Payments

- Staff can record membership payments by selecting a membership, entering the payment amount and method, and recording payment status.
- The system inserts a record into the Payments table and updates the Membership status if necessary.

Track Attendance

- Front desk staff may check in a member by entering the member ID.
- The system will verify active membership and record a check-in timestamp.

- If membership is expired, the system denies entry and displays a notification.

Entities, Attributes, Relationships

Entities:

Members

Members is the basis for all the gym members in the application. Each member is identified by their primary key which is a unique member_id. Additionally, they will also have basic attributes such as phone_number, first_name, last_name, username, email, password_hash, date_joined, and status.

Staff

Staff represents the employees and administrators managing the gym. Each staff member is identified by their primary key which is a unique staff_id. Additionally, they have basic attributes such as first_name, last_name, username, password_hash, role, email, and status.

Trainer

Trainer is a subclass of Staff and is used to represent the staff members acting as trainers. Because Trainer is a subclass of Staff, it inherits its primary key, staff_id, from the Staff table. It also has the attributes of specialization and certification.

Membership Plan

The Membership Plan entity represents the available gym membership tiers. It is identified by its primary key, plan_id, and has the attributes of plan_name, duration_months, price, description, and is_active.

Membership

The Membership entity is used to track a member's selected plan and their current subscription status. It is identified by its primary key, membership_id. It contains important information such as start_date, end_date, status, and freeze_flag, along with the foreign keys member_id and plan_id.

Payment

The Payment entity is used by Members to make transactions for their Memberships. It is

identified by its primary key, payment_id. It contains important information regarding the transaction such as amount, payment_method, payment_date, and payment_status, along with the foreign keys membership_id and member_id.

Class

The Class entity represents the class offerings available at the gym. It is identified by its primary key, class_id. It contains attributes regarding the schedule such as class_name, description, max_capacity, schedule_date, start_time, end_time, and status, along with the foreign key staff_id.

Class Enrollment

Class Enrollment is an associative entity representing a member's enrollment in a class. It is identified by its primary key, enrollment_id. It has the attributes of enrollment_date, enrollment_status, and waitlist_flag, along with the foreign keys member_id and class_id.

AttendanceLog

AttendanceLog is an associative entity storing member check-ins and check-outs at the gym. It is identified by its primary key, attendance_id. It has the attributes of check_in_time and check_out_time, along with the foreign keys member_id and class_id.

Gym

The Gym entity represents the physical gym locations. It is identified by its primary key, gym_name, and has the attributes of addr and hours.

Relationships:

Owned_By

Owned_By represents the relationship between the entities Members and Membership. Specifically, one Member can own many Memberships making the relationship one-to-many where a Membership must be owned by exactly one Member. In the relationship, it uses the PK's member_id and membership_id.

Based_On

Based_On represents the relationship between the entities Membership Plan and Membership. Specifically, one Membership Plan can have many Memberships based on it making the relationship one-to-many where a Membership must be based on exactly one Membership Plan.

In the relationship, it uses the PK's plan_id and membership_id.

Makes_Payment

Makes_Payment represents the relationship between the entities Members and Payment. Specifically, one Member can make many Payments making the relationship one-to-many where a Payment must be made by exactly one Member. In the relationship, it uses the PK's member_id and payment_id.

Pays_For

Pays_For represents the relationship between the entities Membership and Payment. Specifically, one Membership can receive many Payments making the relationship one-to-many where a Payment must pay for exactly one Membership. In the relationship, it uses the PK's membership_id and payment_id.

Led_By

Led_By represents the relationship between the entities Trainer and Class. Specifically, one Trainer can lead many Classes making the relationship one-to-many where a Class must be led by at most one Trainer. In the relationship, it uses the PK's staff_id and class_id.

Enrolls_Member

Enrolls_Member represents the relationship between the entities Members and Class Enrollment. Specifically, one Member can have many Class Enrollments making the relationship one-to-many where a Class Enrollment must enroll exactly one Member. In the relationship, it uses the PK's member_id and enrollment_id.

Enrolls_In_Class

Enrolls_In_Class represents the relationship between the entities Class and Class Enrollment. Specifically, one Class can have many Class Enrollments making the relationship one-to-many where a Class Enrollment must be for exactly one Class. In the relationship, it uses the PK's class_id and enrollment_id.

Member_Log

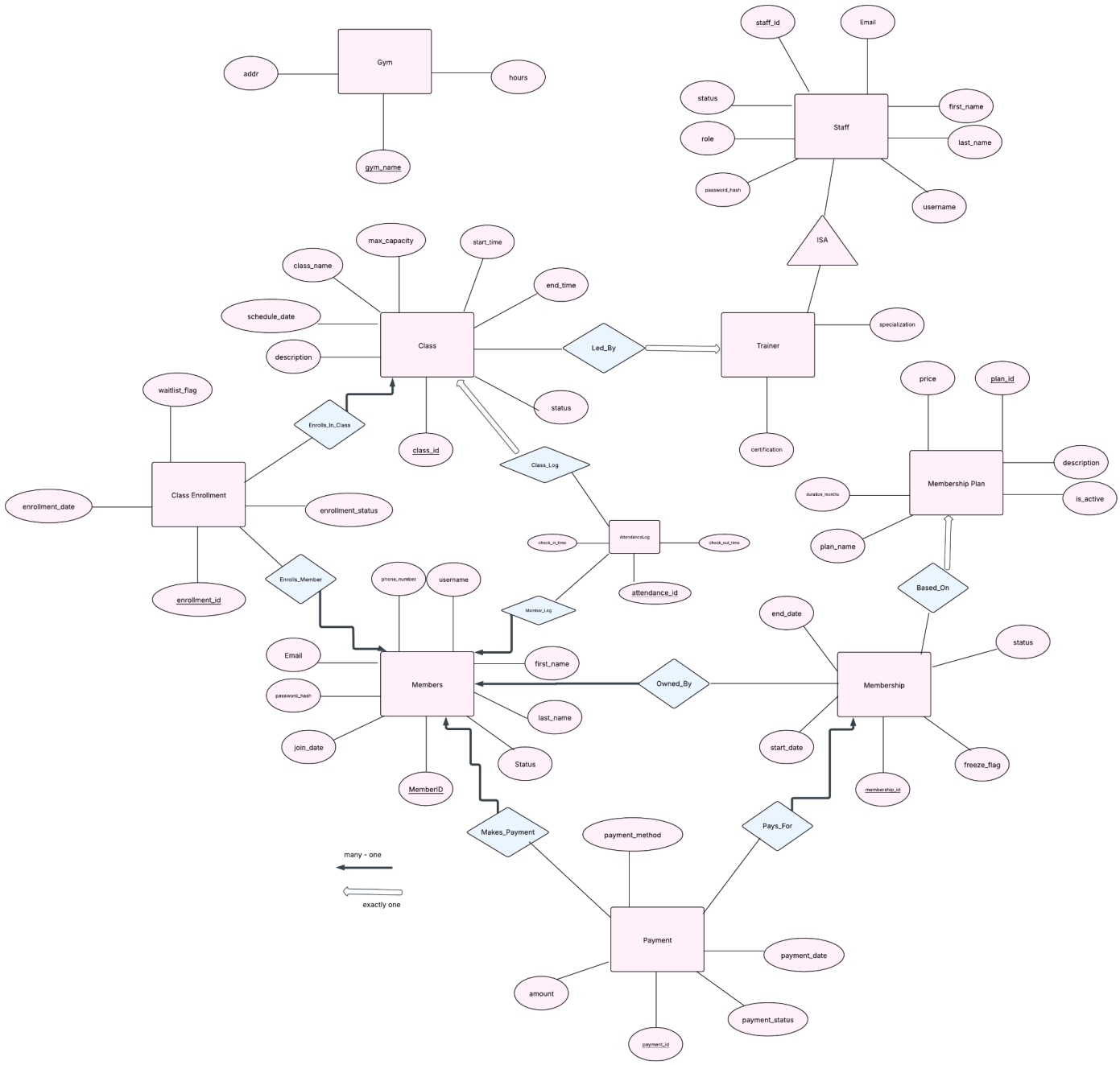
Member_Log represents the relationship between the entities Members and AttendanceLog. Specifically, one Member can have many AttendanceLogs making the relationship one-to-many where an AttendanceLog must belong to exactly one Member. In the relationship, it uses the

PK's member_id and attendance_id.

Class_Log

Class_Log represents the relationship between the entities Class and AttendanceLog. Specifically, one Class can have many AttendanceLogs making the relationship one-to-many where an AttendanceLog must log at most one Class. In the relationship, it uses the PK's class_id and attendance_id.

ERD:
https://lucid.app/lucidchart/4d2adb30-e388-4068-bc9f-89c936194a32/edit?viewport_loc=1202%2C-2061%2C5743%2C3102%2C0_0&invitationId=inv_412f8c62-815d-4b8b-b712-2faf95c176d7


Gym
addr
hours
gym_name
Staff
staff_id
Email
status
first_name
role
last_name
password_hash
username
ISA
Trainer
specialization
certification
Class
max_capacity
start_time
class_name
end_time
schedule_date
description
class_id
status
Led_By
Enrolls_In_Class
Class_Log
Class Enrollment
waitlist_flag
enrollment_date
enrollment_status
enrollment_id
Enrolls_Member
AttendanceLog
check_in_time
check_out_time
attendance_id
Member_Log
Membership Plan
price
plan_id
description
duration_months
is_active
plan_name
Based_On
Members
phone_number
username
Email
password_hash
first_name
last_name
join_date
Status
MemberID
Owned_By
Membership
end_date
status
start_date
freeze_flag
membership_id
Makes_Payment
Pays_For
payment_method
many - one
exactly one
Payment
payment_date
amount
payment_status
payment_id

Relational Schemas

Entities:

Members(member ID, phone number, first_name, last_name, username, email, password_hash, date_joined, status)

Staff(staff_id, first_name, last_name, username, password_hash, role, email, status)

Trainer(*staff_id(PK and FK)*, specialization, certification)

MembershipPlan(plan_id, plan_name, duration_months, price, description, is_active)

Membership(membership_id, *member_id(FK), plan_id(FK)*, start_date, end_date, status, freeze_flag)

Payment(payment_id, *membership_id(FK), member_id(FK)*, amount, payment_method, payment_date, payment_status)

Class(class_id , *staff_id(FK)*, class_name, description, max_capacity, schedule_date, start_time, end_time, status)

ClassEnrollment(enrollment_id, *member_id(FK), class_id(FK)*, enrollment_date, enrollment_status, waitlist_flag)

AttendanceLog(attendance_id, *member_id(FK), class_id(FK)* (nullable), check_in_time, check_out_time)

Gym(gym_name, addr, hours)

MySQL Workbench Tables

Members:

Result Grid Filter Rows: Search Edit: Export/Import:

member_id	phone_number	first_name	last_name	username	email	password_hash	date_join...	status
103	555-0103	Frank	Ocean	focean	frank@email.com	pass3	2026-03-01	Active
104	555-0104	Grace	Hopper	ghopper	grace@email.com	pass4	2026-03-05	Active
105	555-0105	Hank	Pym	hpym	hank@email.com	pass5	2026-03-10	Active
106	555-0106	Iris	West	iwest	iris@email.com	pass6	2026-03-12	Active
107	555-0107	John	Doe	jdoe	john@email.com	pass7	2026-03-15	Frozen
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Staff:

Result Grid Filter Rows: Search Edit: Export/Import:

staff_id	first_name	last_name	username	password_hash	role	email	status
4	Diana	Prince	dprince	hash001	Admin	diana@fithub.com	Active
5	Clark	Kent	ckent	hash002	Trainer	clark@fithub.com	Active
6	Bruce	Wayne	bwayne	hash003	Trainer	bruce@fithub.com	Active
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Trainer:

Result Grid Filter Rows: Search Edit: Export/Import:

staff_id	specialization	certificati...
5	CrossFit	CF-L1
6	Martial Arts	BB-1st
NULL	NULL	NULL

Membership Plan:

Result Grid Filter Rows: Search Edit: Export/Import:

plan_id	plan_name	duration_mont...	price	description	is_active
3	Student Semester	4	100	Discounted student access	Yes
4	VIP Access	12	500	All access + personal locker	Yes
NULL	NULL	NULL	NULL	NULL	NULL

Payment:

Result Grid Filter Rows: Search Edit: Export/Import:

payment_id	membership_id	member_id	amount	payment_meth...	payment_date	payment_stat...
503	1003	103	100	Credit Card	2026-03-01	Completed
504	1004	104	500	Credit Card	2026-03-05	Completed
505	1005	105	30	Cash	2026-03-10	Completed
506	1006	106	300	Debit Card	2026-03-12	Completed
507	1007	107	30	Cash	2026-03-15	Completed
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Class:

Result Grid Filter Rows: Search Edit: Export/Import:

class_id	class_name	description	max_capacity	schedule_date	start_time	end_time	staff_id	status
203	HIIT Blast	High intensity interval training	20	2026-03-22	12:00	13:00	5	Scheduled
204	Kickboxing	Cardio kickboxing	15	2026-03-23	18:00	19:00	6	Scheduled
205	Evening Yoga	Wind down stretch	25	2026-03-24	20:00	21:00	3	Scheduled
206	Powerlifting	Advanced bar techniques	10	2026-03-25	16:00	17:30	2	Scheduled
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Class Enrollment

Result Grid Filter Rows: Search Edit: Export/Import:

enrollment_id	member_id	class_id	enrollment_da...	enrollment_stat...	waitlist_fl...
303	103	203	2026-03-20	Enrolled	No
304	104	204	2026-03-20	Enrolled	No
305	105	205	2026-03-21	Enrolled	No
306	106	206	2026-03-21	Enrolled	No
307	101	203	2026-03-21	Waitlisted	Yes
308	102	204	2026-03-21	Enrolled	No
NULL	NULL	NULL	NULL	NULL	NULL

Attendance Log:

Result Grid Filter Rows: Search Edit: Export/Import:

attendance_id	member_id	class_id	check_in_time	check_out_time
403	103	203	2026-03-22 11:50	2026-03-22 13:10
404	104	204	2026-03-23 17:55	2026-03-23 19:15
405	105	NULL	2026-03-20 09:00	2026-03-20 10:30
406	106	NULL	2026-03-21 14:00	2026-03-21 15:45
407	101	NULL	2026-03-22 08:00	2026-03-22 09:30
408	102	201	2026-03-20 07:50	2026-03-20 09:10
NULL	NULL	NULL	NULL	NULL

Gym:

Result Grid Filter Rows: Search Edit: Export/Import:

gym_name	addr	hours
Northside FitHub	789 North Blvd, San Jose, CA	05:00-23:00
NULL	NULL	NULL